Lista 03 — DISTINCT, AND-OR-NOT (Hands-On)

SQL para Estatísticos — Laboratório 03

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Preparação

O banco `me315.sqlite` contém as tabelas `PRESERVE` e `EMPLOYEE` criadas na aula anterior. Para que este documento seja reproduzível, o banco é (re)criado abaixo com os mesmos comandos da Aula 01, e a conexão `con` é usada em todos os chunks SQL.

```
library(DBI)
library(RSQLite)

con <- dbConnect(SQLite(), "me315.sqlite")

# Mostrar todas as linhas nos chunks SQL (o padrão do knitr é 10)
knitr::opts_chunk$set(connection = "con", max.print = -1)

comandos <- c(
  "DROP TABLE IF EXISTS PRESERVE",
  "DROP TABLE IF EXISTS EMPLOYEE",
  "CREATE TABLE PRESERVE
   (PNO   INTEGER       NOT NULL UNIQUE,
    PNAME VARCHAR (25)  NOT NULL,
    STATE CHAR (2)      NOT NULL,
    ACRES INTEGER       NOT NULL,
    FEE   DECIMAL (5,2) NOT NULL)",
  "INSERT INTO PRESERVE VALUES
   (5,  'HASSAYAMPA RIVER',        'AZ', 660,   3.00),
   (3,  'DANCING PRAIRIE',         'MT', 680,   0.00),
   (7,  'MULESHOE RANCH',          'AZ', 49120, 0.00),
   (40, 'SOUTH FORK MADISON',      'MT', 121,   0.00),
   (14, 'MCELWAIN-OLSEN',          'MA', 66,    0.00),
   (13, 'TATKON',                  'MA', 40,    0.00),
   (9,  'DAVID H. SMITH',          'MA', 830,   0.00),
   (11, 'MIACOMET MOORS',          'MA', 4,     0.00),
   (12, 'MOUNT PLANTAIN',          'MA', 730,   0.00),
   (1,  'COMERTOWN PRAIRIE',       'MT', 1130,  0.00),
   (2,  'PINE BUTTE SWAMP',        'MT', 15000, 0.00),
   (80, 'RAMSEY CANYON',           'AZ', 380,   3.00),
   (10, 'HOFT FARM',               'MA', 90,    0.00),
   (6,  'PAPAGONIA-SONOITA CREEK', 'AZ', 1200,  3.00)",
  "CREATE TABLE EMPLOYEE
   (ENO    CHAR (4)      NOT NULL UNIQUE,
```

```
    ENAME  VARCHAR(25)   NOT NULL,
    SALARY DECIMAL(7,2)  NOT NULL,
    DNO    INTEGER       NOT NULL)",
  "INSERT INTO EMPLOYEE VALUES
   ('1000', 'MOE',    2000.00, 20),
   ('2000', 'LARRY',  2000.00, 10),
   ('3000', 'CURLY',  3000.00, 20),
   ('4000', 'SHEMP',   500.00, 40),
   ('5000', 'JOE',     400.00, 10),
   ('6000', 'GEORGE', 9000.00, 20)"
)
invisible(lapply(comandos, function(q) dbExecute(con, q)))

dbListTables(con)
```

```
[1] "EMPLOYEE" "PRESERVE"
```

DISTINCT

Exercício 1

Exiba os valores de `STATE` e `FEE` da tabela `PRESERVE`, mas sem duplicados.

```
SELECT DISTINCT STATE, FEE
FROM PRESERVE;
```

Tabela 1: 4 records

STATE	FEE
AZ	3
MT	0
AZ	0
MA	0

Exercício 2

Exiba todos os valores de `DNO` na tabela `EMPLOYEE`. Não exiba valores duplicados.

```
SELECT DISTINCT DNO
FROM EMPLOYEE;
```

Tabela 2: 3 records

DNO
20
10
40

Exercício 3

Execute as consultas a seguir. Examine as tabelas resultantes e faça as observações relevantes.

```
SELECT DNO, SALARY
FROM EMPLOYEE;
```

Tabela 3: 6 records

DNO	SALARY
20	2000
10	2000
20	3000
40	500
10	400
20	9000

```
SELECT DISTINCT DNO, SALARY
FROM EMPLOYEE;
```

Tabela 4: 6 records

DNO	SALARY
20	2000
10	2000
20	3000
40	500
10	400
20	9000

```
SELECT DISTINCT DNO, SALARY
FROM EMPLOYEE
ORDER BY DNO, SALARY;
```

Tabela 5: 6 records

DNO	SALARY
10	400
10	2000
20	2000
20	3000
20	9000
40	500

AND, OR e NOT

Exercício 4

Exiba todas as informações sobre qualquer reserva natural em Montana que seja menor que 1.000 acres.

```
SELECT *
FROM PRESERVE
WHERE STATE = 'MT' AND ACRES < 1000;
```

Tabela 6: 2 records

PNO	PNAME	STATE	ACRES	FEE
3	DANCING PRAIRIE	MT	680	0
40	SOUTH FORK MADISON	MT	121	0

Exercício 5

Exiba todas as informações sobre qualquer reserva natural cujo valor de `ACRES` esteja entre 1.200 e 15.000, inclusive.

```
SELECT *
FROM PRESERVE
WHERE ACRES >= 1200 AND ACRES <= 15000;
```

Tabela 7: 2 records

PNO	PNAME	STATE	ACRES	FEE
2	PINE BUTTE SWAMP	MT	15000	0
6	PAPAGONIA-SONOITA CREEK	AZ	1200	3

Exercício 6

Exiba todas as informações sobre qualquer reserva natural localizada em Montana que não possua taxa de entrada e seja maior que 10.000 acres.

```
SELECT *
FROM PRESERVE
WHERE STATE = 'MT' AND FEE = 0 AND ACRES > 10000;
```

Tabela 8: 1 records

PNO	PNAME	STATE	ACRES	FEE
2	PINE BUTTE SWAMP	MT	15000	0

Exercício 7

Exiba todas as informações sobre as reservas naturais localizadas em Montana ou Massachusetts.

```sql
SELECT *
FROM PRESERVE
WHERE STATE = 'MT' OR STATE = 'MA';
```

Tabela 9: 10 records

PNO	PNAME	STATE	ACRES	FEE
3	DANCING PRAIRIE	MT	680	0
40	SOUTH FORK MADISON	MT	121	0
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
9	DAVID H. SMITH	MA	830	0
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0
10	HOFT FARM	MA	90	0

Exercício 8

Exiba todas as informações sobre qualquer reserva natural localizada em Montana ou qualquer reserva que seja menor que 1.000 acres.

```sql
SELECT *
FROM PRESERVE
WHERE STATE = 'MT' OR ACRES < 1000;
```

Tabela 10: 12 records

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3
3	DANCING PRAIRIE	MT	680	0
40	SOUTH FORK MADISON	MT	121	0
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
9	DAVID H. SMITH	MA	830	0
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0
80	RAMSEY CANYON	AZ	380	3
10	HOFT FARM	MA	90	0

Exercício 9

Exiba o número e o nome de qualquer reserva natural que possua taxa de entrada que não seja igual a zero.

```sql
SELECT PNO, PNAME
FROM PRESERVE
```

```
WHERE FEE <> 0;
```

Tabela 11: 3 records

PNO	PNAME
5	HASSAYAMPA RIVER
80	RAMSEY CANYON
6	PAPAGONIA-SONOITA CREEK

Exercício 10

Exiba o número e o nome das reservas naturais que não possuem taxa de entrada de US$ 3,00 e não possuem taxa de US$ 10,00.

```
SELECT PNO, PNAME
FROM PRESERVE
WHERE NOT FEE = 3 AND NOT FEE = 10;
```

Tabela 12: 11 records

PNO	PNAME
3	DANCING PRAIRIE
7	MULESHOE RANCH
40	SOUTH FORK MADISON
14	MCELWAIN-OLSEN
13	TATKON
9	DAVID H. SMITH
11	MIACOMET MOORS
12	MOUNT PLANTAIN
1	COMERTOWN PRAIRIE
2	PINE BUTTE SWAMP
10	HOFT FARM

Exercício 11

Exiba todas as informações sobre qualquer reserva natural localizada no Arizona que não possua taxa de entrada, ou qualquer reserva que seja menor que 100 acres (independentemente de seus valores de `STATE` e `FEE`).

```
SELECT *
FROM PRESERVE
WHERE (STATE = 'AZ' AND FEE = 0) OR ACRES < 100;
```

Tabela 13: 5 records

PNO	PNAME	STATE	ACRES	FEE
7	MULESHOE RANCH	AZ	49120	0
14	MCELWAIN-OLSEN	MA	66	0

PNO	PNAME	STATE	ACRES	FEE
13	TATKON	MA	40	0
11	MIACOMET MOORS	MA	4	0
10	HOFT FARM	MA	90	0

Exercício 12

Exiba todas as informações sobre qualquer reserva natural que seja menor que 1.000 acres e tenha taxa de entrada de zero dólares ou esteja localizada no Arizona.

```
SELECT *
FROM PRESERVE
WHERE ACRES < 1000 AND (FEE = 0 OR STATE = 'AZ');
```

Tabela 14: 10 records

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3
3	DANCING PRAIRIE	MT	680	0
40	SOUTH FORK MADISON	MT	121	0
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
9	DAVID H. SMITH	MA	830	0
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0
80	RAMSEY CANYON	AZ	380	3
10	HOFT FARM	MA	90	0

Exercício 13

Selecione todas as informações sobre qualquer reserva natural com taxa de entrada que não seja maior que zero, ou qualquer outra reserva, independentemente de sua taxa, que esteja localizada em Montana e seja maior que 1.000 acres.

```
SELECT *
FROM PRESERVE
WHERE NOT FEE > 0 OR (STATE = 'MT' AND ACRES > 1000);
```

Tabela 15: 11 records

PNO	PNAME	STATE	ACRES	FEE
3	DANCING PRAIRIE	MT	680	0
7	MULESHOE RANCH	AZ	49120	0
40	SOUTH FORK MADISON	MT	121	0
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
9	DAVID H. SMITH	MA	830	0
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0

PNO	PNAME	STATE	ACRES	FEE
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0
10	HOFT FARM	MA	90	0

Exercício 14

Exiba todas as informações sobre todas as reservas naturais, exceto aquelas reservas de Montana sem taxa de entrada.

```
SELECT *
FROM PRESERVE
WHERE NOT (STATE = 'MT' AND FEE = 0);
```

Tabela 16: 10 records

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3
7	MULESHOE RANCH	AZ	49120	0
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
9	DAVID H. SMITH	MA	830	0
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0
80	RAMSEY CANYON	AZ	380	3
10	HOFT FARM	MA	90	0
6	PAPAGONIA-SONOITA CREEK	AZ	1200	3

Equivalência lógica de cláusulas WHERE

Exercício 15

As seguintes cláusulas `WHERE` são logicamente equivalentes?

- `WHERE STATE = 'MA' AND (ACRES > 1000 OR FEE = 0.0)`
- `WHERE (STATE = 'MA' AND ACRES > 1000) OR (STATE = 'MA' AND FEE = 0.0)`

```
SELECT *
FROM PRESERVE
WHERE STATE = 'MA' AND (ACRES > 1000 OR FEE = 0.0);
```

Tabela 17: 6 records

PNO	PNAME	STATE	ACRES	FEE
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
9	DAVID H. SMITH	MA	830	0

PNO	PNAME	STATE	ACRES	FEE
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0
10	HOFT FARM	MA	90	0

```
SELECT *
FROM PRESERVE
WHERE (STATE = 'MA' AND ACRES > 1000) OR (STATE = 'MA' AND FEE = 0.0);
```

Tabela 18: 6 records

PNO	PNAME	STATE	ACRES	FEE
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
9	DAVID H. SMITH	MA	830	0
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0
10	HOFT FARM	MA	90	0

Exercício 16

As seguintes cláusulas `WHERE` são logicamente equivalentes?

- `WHERE STATE = 'MA' OR (ACRES > 1000 AND FEE = 0.0)`
- `WHERE (STATE = 'MA' OR ACRES > 1000) AND (STATE = 'MA' OR FEE = 0.0)`

```
SELECT *
FROM PRESERVE
WHERE STATE = 'MA' OR (ACRES > 1000 AND FEE = 0.0);
```

Tabela 19: 9 records

PNO	PNAME	STATE	ACRES	FEE
7	MULESHOE RANCH	AZ	49120	0
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
9	DAVID H. SMITH	MA	830	0
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0
10	HOFT FARM	MA	90	0

```
SELECT *
FROM PRESERVE
WHERE (STATE = 'MA' OR ACRES > 1000) AND (STATE = 'MA' OR FEE = 0.0);
```

Tabela 20: 9 records

PNO	PNAME	STATE	ACRES	FEE
7	MULESHOE RANCH	AZ	49120	0
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
9	DAVID H. SMITH	MA	830	0
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0
10	HOFT FARM	MA	90	0

Exercício 17

As seguintes cláusulas `WHERE` são logicamente equivalentes?

- `WHERE NOT (ACRES < 50 AND STATE = 'MA')`
- `WHERE NOT ACRES < 50 AND NOT STATE = 'MA'`
- `WHERE ACRES >= 50 OR STATE <> 'MA'`

```
SELECT *
FROM PRESERVE
WHERE NOT (ACRES < 50 AND STATE = 'MA');
```

Tabela 21: 12 records

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3
3	DANCING PRAIRIE	MT	680	0
7	MULESHOE RANCH	AZ	49120	0
40	SOUTH FORK MADISON	MT	121	0
14	MCELWAIN-OLSEN	MA	66	0
9	DAVID H. SMITH	MA	830	0
12	MOUNT PLANTAIN	MA	730	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0
80	RAMSEY CANYON	AZ	380	3
10	HOFT FARM	MA	90	0
6	PAPAGONIA-SONOITA CREEK	AZ	1200	3

```
SELECT *
FROM PRESERVE
WHERE NOT ACRES < 50 AND NOT STATE = 'MA';
```

Tabela 22: 8 records

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3
3	DANCING PRAIRIE	MT	680	0
7	MULESHOE RANCH	AZ	49120	0
40	SOUTH FORK MADISON	MT	121	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0
80	RAMSEY CANYON	AZ	380	3
6	PAPAGONIA-SONOITA CREEK	AZ	1200	3

```sql
SELECT *
FROM PRESERVE
WHERE ACRES >= 50 OR STATE <> 'MA';
```

Tabela 23: 12 records

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3
3	DANCING PRAIRIE	MT	680	0
7	MULESHOE RANCH	AZ	49120	0
40	SOUTH FORK MADISON	MT	121	0
14	MCELWAIN-OLSEN	MA	66	0
9	DAVID H. SMITH	MA	830	0
12	MOUNT PLANTAIN	MA	730	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0
80	RAMSEY CANYON	AZ	380	3
10	HOFT FARM	MA	90	0
6	PAPAGONIA-SONOITA CREEK	AZ	1200	3

Exercício 18

As seguintes cláusulas `WHERE` são logicamente equivalentes?

- `WHERE NOT (ACRES < 50 OR STATE = 'MA')`
- `WHERE NOT ACRES < 50 OR NOT STATE = 'MA'`
- `WHERE NOT ACRES < 50 AND NOT STATE = 'MA'`

```sql
SELECT *
FROM PRESERVE
WHERE NOT (ACRES < 50 OR STATE = 'MA');
```

Tabela 24: 8 records

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3

PNO	PNAME	STATE	ACRES	FEE
3	DANCING PRAIRIE	MT	680	0
7	MULESHOE RANCH	AZ	49120	0
40	SOUTH FORK MADISON	MT	121	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0
80	RAMSEY CANYON	AZ	380	3
6	PAPAGONIA-SONOITA CREEK	AZ	1200	3

```
SELECT *
FROM PRESERVE
WHERE NOT ACRES < 50 OR NOT STATE = 'MA';
```

Tabela 25: 12 records

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3
3	DANCING PRAIRIE	MT	680	0
7	MULESHOE RANCH	AZ	49120	0
40	SOUTH FORK MADISON	MT	121	0
14	MCELWAIN-OLSEN	MA	66	0
9	DAVID H. SMITH	MA	830	0
12	MOUNT PLANTAIN	MA	730	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0
80	RAMSEY CANYON	AZ	380	3
10	HOFT FARM	MA	90	0
6	PAPAGONIA-SONOITA CREEK	AZ	1200	3

```
SELECT *
FROM PRESERVE
WHERE NOT ACRES < 50 AND NOT STATE = 'MA';
```

Tabela 26: 8 records

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3
3	DANCING PRAIRIE	MT	680	0
7	MULESHOE RANCH	AZ	49120	0
40	SOUTH FORK MADISON	MT	121	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0
80	RAMSEY CANYON	AZ	380	3
6	PAPAGONIA-SONOITA CREEK	AZ	1200	3